

Ex. No 8: Detect Hidden Data in Images Using StegExpose

Date:

Aim

To detect hidden data in image files using StegExpose, a steganography analysis tool.

Procedure

Step 1: Compile the Source Code

Compile the Java source files with the required dependency:

```
javac -cp commons-math3-3.1.1.jar -source 1.8 -target 1.8 *.java
```

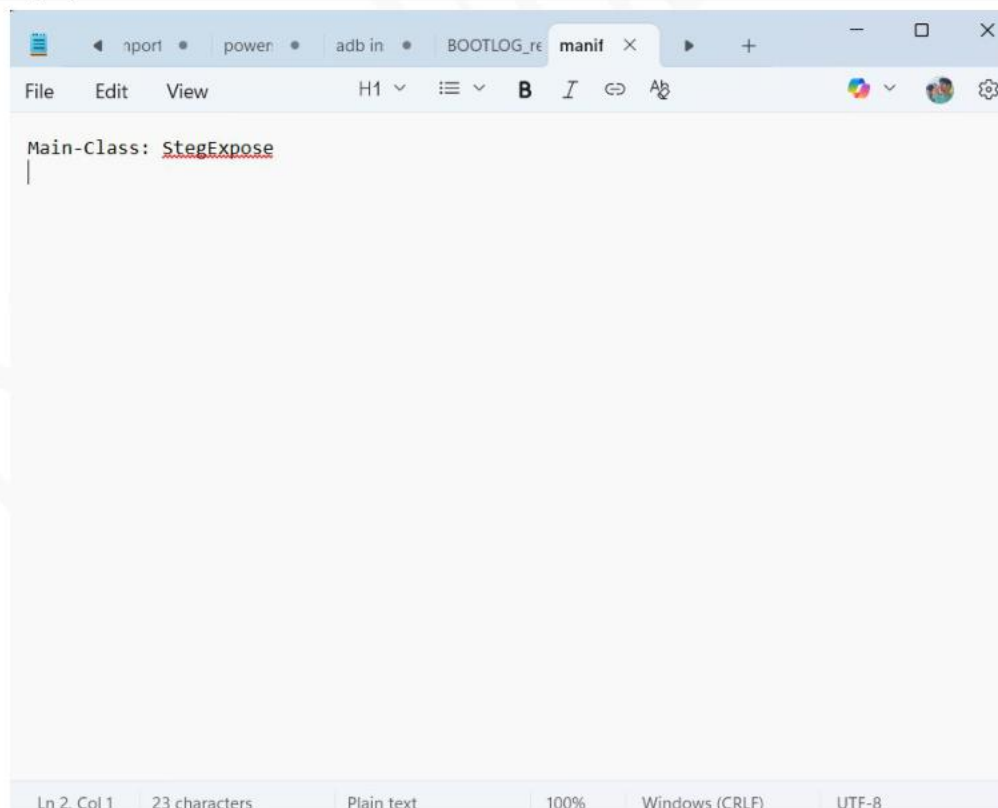
```
warning: [options] bootstrap class path is not set in conjunction with -source 8
not setting the bootstrap class path may lead to class files that cannot run on JDK 8
--release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
warning: [options] source value 8 is obsolete and will be removed in a future release
warning: [options] target value 8 is obsolete and will be removed in a future release
warning: [options] To suppress warnings about obsolete options, use -Xlint:-options.
Note: RSAnalysis.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.
4 warnings
```

Note: Ignore warnings like 'RSAnalysis.java uses unchecked or unsafe operations'. Compilation is successful if .class files are generated.

Step 2: Create the Executable JAR

Create a manifest file named manifest.mf in the project folder:

Main-Class: StegExpose



Build the standalone executable JAR:

```
jar cfm StegExpose.jar manifest.mf *.class
```

Step 3: Run StegExpose

Execute StegExpose against the test image directory:

```
java -jar StegExpose.jar "E:\StegExpose-master\StegExpose-master\testFolder"
```

Step 4: Output Analysis & Steganographic Thresholds

```
se-master\StegExpose-master\testFolder"
stego_6666458261_e455d262b5_z.png is suspicious. Approximate amount of hidden data is 114785 bytes.
stego_6672108499_85c582a7f9.png is suspicious. Approximate amount of hidden data is 137047 bytes.
stego_6672542201_532f70bffe.png is suspicious. Approximate amount of hidden data is 67141 bytes.
```

StegExpose reports potential hidden payload estimates according to established statistical thresholds:

Hidden Bytes	Diagnostic Meaning
< 10,000	Likely clean (no hidden data detected)
10,000 – 100,000	Suspicious (possibly contains hidden steganographic data)
> 100,000	Highly likely steganography present (requires manual verification)

Step 5: Analyze and Verify Results

The tool lists images with potential hidden data and estimates the approximate size. Further validation can be done using tools such as:

- StegSolve (Bit-plane visualization)
- OpenStego (Payload extraction)
- zsteg (LSB PNG/BMP analysis)
- Binwalk (Firmware and appended payload analysis)

Step 6: Export Results Automatically

Generate an automated CSV results report for documentation:

```
java -jar StegExpose.jar "E:\StegExpose-master\StegExpose-master\testFolder" fast 0.3 results.csv
```

```
C:\Users\knsha\Downloads\StegExpose-master\StegExpose-master>java -jar StegExpose.jar "C:\Users\knsha\Downloads\StegExpose-master\StegExpose-master\testFolder"
stego_6666458261_e455d262b5_z.png is suspicious. Approximate amount of hidden data is 114785 bytes.
stego_6672108499_85c582a7f9.png is suspicious. Approximate amount of hidden data is 137047 bytes.
stego_6672542201_532f70bffe.png is suspicious. Approximate amount of hidden data is 67141 bytes.

C:\Users\venkat_jyothibabu\Downloads\StegExpose-master\StegExpose-master>java -jar StegExpose.jar "C:\Users\venkat_jyothibabu\Downloads\StegExpose-master\StegExpose-master\testFolder" fast 0.3 results.csv

C:\Users\venkat_jyothibabu\Downloads\StegExpose-master\StegExpose-master>
```

Reviewing exported CSV findings:

File name	Above stego threshold?	Secret message size in bytes (ignore for clean files)	Primary Sets	Chi Square	Sample Pairs	RS analysis:Fusion (mean & fast)
clean_7221386570_82a10bc040_z.png	FALSE		14980	0.113292402	null	0.113292402
clean_7222762386_2688184c8e.png	FALSE		10675	NaN	0.240066026	0.240066026
clean_7228374452_8f743dc3d4_z.png	FALSE		4192	0.031137469	null	0.031137469
clean_7228718722_1cf25d0ff3e_z.png	FALSE		22673	0.116140906	null	0.116140906
clean_7231623538_f51db2a35a_z.png	FALSE		9527	0.050937673	null	0.050937673
clean_7232193260_zf08b0f25_z.png	FALSE		50109	0.257388306	null	0.257388306
clean_7232202430_d38b6d6986_z.png	FALSE		33951	NaN	0.180170577	0.180170577
clean_7232206610_b8cfed120_z.png	FALSE		5061	0.028480431	null	0.028480431
clean_723220662_3d42c89109_z.png	FALSE		24505	NaN	0.13287066	0.13287066
clean_7234473324_b0ba82b05d.png	FALSE		1261	0.016539926	null	0.016539926
clean_7235558256_3099066753.png	FALSE		20378	0.175461027	null	0.175461027
clean_7235972310_8c25258da5.png	FALSE		21010	0.184964588	null	0.184964588
clean_7237798856_86870c5823_z.png	FALSE		5587	0.073302064	null	0.073302064
stego_6666458261_e455d262b5_z.png	TRUE		114785	NaN	0.067670212	0.729290221 0.738013
stego_6672108499_85c582a7f9.png	TRUE		137047	NaN	0.999990598	0.869877 0.934933814
stego_6672542201_532f70bffe.png	TRUE		67141	NaN	0.09371292	0.781241535 0.768072

Rubrics

Criteria	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

- Successfully compiled and executed the StegExpose steganography detection suite.
- Analyzed image test sets using Chi-square, Primary Sets, Sample Pair, and RS analysis algorithms.
- Detected images containing potential hidden payloads, interpreted threshold metrics, and exported findings to CSV.